

MSE 360: Kinetic Processes in Materials
S2023
Midterm Exam 2

Points: ____/100

Name (print): _____

I have neither given nor received unauthorized aid on this test.

_____ (signature)

No notes, books, cell phones, or information stored in calculator memories may be used. Suspicion of cheating will result in the filing of a Report of Academic Integrity Violation. All of your work must be written on these pages and turned in.

Useful Constants

$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$	$1 \text{ eV/atom} = 23.06 \text{ kcal/mol}$
$R = 1.987 \text{ cal mol}^{-1} \text{ K}^{-1}$	$k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$
$R = 8.3142 \text{ J mol}^{-1} \text{ K}^{-1}$	$k_B = 8.616 \times 10^{-5} \text{ eV K}^{-1}$
$R = 82.05 \text{ cm}^3 \text{ atm mol}^{-1} \text{ K}^{-1}$	$N_A = 6.022 \times 10^{23} \text{ molecules mol}^{-1}$
$e = 1.602 \times 10^{-19} \text{ C}$	$101,325 \text{ Pa} = 1 \text{ atm}$

ω	$\text{erf}[\omega]$	ω	$\text{erf}[\omega]$	ω	$\text{erf}[\omega]$	ω	$\text{erf}[\omega]$
0.00	0.000000	0.40	0.428392	0.80	0.742101	1.40	0.952285
0.02	0.022565	0.42	0.447468	0.82	0.753811	1.44	0.958297
0.04	0.045111	0.44	0.466225	0.84	0.765143	1.48	0.963654
0.06	0.067622	0.46	0.484655	0.86	0.776100	1.52	0.968413
0.08	0.090078	0.48	0.502750	0.88	0.786687	1.56	0.972628
0.10	0.112463	0.50	0.520500	0.90	0.796908	1.60	0.976348
0.12	0.134758	0.52	0.537899	0.92	0.806768	1.70	0.983790
0.14	0.156947	0.54	0.554939	0.94	0.816271	1.80	0.989091
0.16	0.179012	0.56	0.571616	0.96	0.825424	1.90	0.992790
0.18	0.200936	0.58	0.587923	0.98	0.834232	2.00	0.995322
0.20	0.222703	0.60	0.603856	1.00	0.842701	2.10	0.997021
0.22	0.244296	0.62	0.619411	1.04	0.858650	2.20	0.998137
0.24	0.265700	0.64	0.634586	1.08	0.873326	2.30	0.998857
0.26	0.286900	0.66	0.649377	1.12	0.886788	2.40	0.999311
0.28	0.307880	0.68	0.663782	1.16	0.899096	2.50	0.999593
0.30	0.328627	0.70	0.677801	1.20	0.910314	2.60	0.999764
0.32	0.349126	0.72	0.691433	1.24	0.920505	2.70	0.999866
0.34	0.369365	0.74	0.704678	1.28	0.929734	2.80	0.999925
0.36	0.389330	0.76	0.717537	1.32	0.938065	2.90	0.999959
0.38	0.409009	0.78	0.730010	1.36	0.945561	3.00	0.999978

Table 4.3: Look-up Table of Error Function Values. To determine corresponding values for the Complementary Error Function, recall that $\text{erfc}[\omega] = 1 - \text{erf}[\omega]$

$\text{erf}[\omega]$	Value	$\text{erfc}[\omega]$	Value
$\text{erf}[-\infty]$	-1	$\text{erfc}[-\infty]$	2
$\text{erf}[0]$	0	$\text{erfc}[0]$	1
$\text{erf}[\infty]$	1	$\text{erfc}[\infty]$	0